

Question for Lab 3_Batch Even

Q1. For the Classes given below give complete implementation and upload all the classes

Ensure that you

- Include all classes, including Library classes.
- All methods and field are implemented
- Access specifiers are provided suitably.
- Also include a class Main which has a main function which will create one object of each of these classes
- Ensure you also add any setters and getters needed for the classes to function as required.

```
import java.util.*;
import java.math.*;
public class Point{
    private int x, y;
    public Point(int x, int y){
        this.x = x;
        this.y = y;
    }
    public void setX(int x){
        this.x = x;
    }
    public void setY(int y){
        this.y = y;
    }
    public int getX(){
        return this.x;
    }
    public int getY(){
        return this.y;
    }
    public double getDistance(Point other){
        return Math.sqrt(Math.pow(this.x - other.x, 2) + Math.pow(this.y
- other.y, 2));
    }
}
```

```
public class Triangle {
    private Point point1, point2, point3;
    public Triangle(int x1, int x2, int x3, int y1, int y2, int y3){
        //Implement constructor given 6 coordinates
    }
    public Triangle(Point point1, Point point2, Point point3){
        //Implement constructor given 3 coordinates
    }
    //Implement getters, setters
    //Implement given methods
    public boolean isIsosceles(){
```

```
        return false;
    }
    public boolean isEquilateral(){
        return false;
    }
    public boolean isCollinear(){
        return false;
    }
    public double getArea(){
        return 0;
    }
}
```
